



3par Host Connectivity



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Host Connectivity

- ▶ The HPE 3PAR StoreServ Storage system sees a host as a set of initiator port World Wide Names (WWN) or iSCSI names.
- ▶ Hosts that are physically connected to ports on the system are automatically detected.
- ▶ The FC port WWNs and iSCSI port iSCSI names are displayed by the user interfaces. You can also add new WWNs or iSCSI names for unestablished host paths and assign them to a host before they are physically connected.



Host Ports

- ▶ FC over Ethernet (FCoE) connectivity is supported on the HPE 3PAR StoreServ Storage systems with the use of converged network adapters (CNA).
- ▶ CNA ports can be configured for use as FCoE or iSCSI ports.
- ▶ A virtual volume can be exported, or made accessible, to one or more hosts.
- ▶ The host sees the exported virtual volume as a LUN connected to one or more ports. After the virtual volume is exported to a host, the host can send requests to the LUN.



Ports

- ▶ Persistent ports (also called virtual ports) allow the HPE 3PAR StoreServ Storage system host-facing ports to assume the identity of partner ports that are automatically designated by the system.
- ▶ For iSCSI, FC, and FCoE ports, this is achieved by using N_Port ID Virtualization (NPIV).



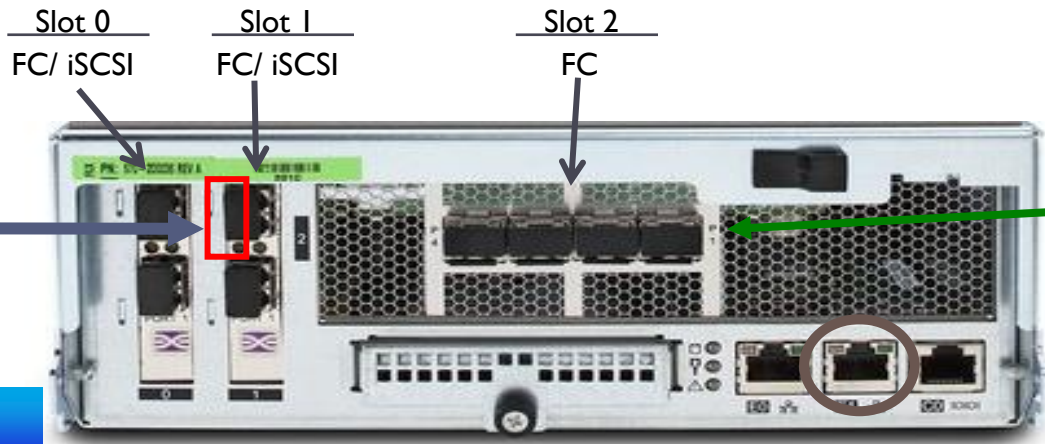
Ports

- ▶ System controller nodes can use FC, Gigabit Ethernet, and iSCSI, and FCoE ports to connect the storage system to your network, host computers, storage system components, and to other systems.
- ▶ Serial Attached SCSI (SAS) ports are only used to connect system components.
 - ▶ **FC ports**
 - ▶ **iSCSI ports**
 - ▶ **Gigabit Ethernet ports**
 - ▶ **SAS ports**
 - ▶ **FCoE Ports**



Fibre Channel and PCI Slot Numbering

➡ Note Numbering Exception: Fibre Channel top starts with “1”



Node 3

Referencing a “port” on a node = Node #:PCI slot:port #

3:1:1



StorServ Numbering Systems

Know the numbering system:

▶ N:S:P

- Nodes
- Slots (PCI-X)
- Ports

General Rule:	Top = Starts with 0
	Left = Starts with 0

▶ C:M:D

- Chassis/Cage
- Magazines
- Disks



dxs-3par cli% showport -par

N:S:P	Connmode	ConnType	CfgRate	MaxRate	Class2	UniqNodeWwn	VCN	IntCoal	TMWO
Smart_SAN									
0:0:1	disk	point	6Gbps	6Gbps	n/a	n/a	n/a	enabled	n/a
n/a									
0:1:1	disk	point	6Gbps	6Gbps	n/a	n/a	n/a	enabled	n/a
n/a									
0:2:1	disk	point	6Gbps	6Gbps	n/a	n/a	n/a	enabled	n/a
n/a									
0:4:1	disk	point	6Gbps	6Gbps	n/a	n/a	n/a	enabled	n/a
n/a									
0:9:1	disk	loop	n/a	8Gbps	n/a	disabled	disabled	enabled	disa
bled n/a									
1:0:1	disk	point	6Gbps	6Gbps	n/a	n/a	n/a	enabled	n/a
n/a									
1:1:1	disk	point	6Gbps	6Gbps	n/a	n/a	n/a	enabled	n/a
n/a									



Port target, initiator, and peer modes

- ▶ The system controller node ports operate in different modes. Depending on the type of port, the port may operate in target, initiator, or peer mode.
- ▶ **FC ports** use the following firmware mode settings:
 - ▶ • Target mode for ports that connect to hosts and receive commands from those hosts.
 - ▶ • Initiator mode for ports that connect to the system physical disks and send commands to those disks.
 - ▶ • Initiator mode for Remote Copy over FC (RCFC).



Ports

- ▶ **iSCSI ports** use the following firmware mode setting:
 - ▶ Target mode for ports that connect to hosts and receive commands from those hosts.
- ▶ **Gigabit Ethernet ports** use the following firmware mode setting:
 - ▶ Peer mode for Ethernet ports, used for RCIP.
- ▶ **FCoE ports** use the following firmware mode setting:
 - ▶ Target mode for ports that connect to hosts and receive commands from those hosts.
- ▶ **SAS ports**
 - ▶ Initiator mode for ports that connect to the system physical disks and send commands to those disks.





Host Explorer

Host Explorer

- ▶ The Host Explorer collects detailed host side storage stack configuration information on the host it is installed and communicates it to HPE 3PAR or HPE Primera storage systems over the front-end FC or iSCSI connection.
- ▶ The host side storage stack information will be used to assists with new Host object creation on the storage systems and for diagnosing host connectivity issues.



Host Explorer

- ▶ The Host Explorer collects the following information and sends it to the connected storage system:
- ▶ Host operating system and version.
- ▶ Fibre Channel and iSCSI HBA details.
- ▶ Multipath driver and current multipath configuration.
- ▶ Cluster configuration information.
- ▶ Device and path Information.
- ▶ Host capacity utilization information



Host Explorer Agent Install



3PAR Host Explorer agent is supported on the following platforms:

- Microsoft Windows Server 2003 and 2008, 32 bit and 64 bit
- Red Hat Linux systems running RHEL4 Update 6 or later and RHEL5
- Sparc systems running Solaris 10



The 3PAR Host Explorer supports VMware Virtual Machines connected to StorServ Storage Servers with iSCSI connections

Host Explorer

Host Explorer Details: DL36-529						
Host Name	DL36-529	WWN	Vendor	Model	Firmware Version	Driver Version
ID	4	5001438003B0B70C	QLogic Corporation	HPAE312A	5.01.03	9.1.8.25
Domain	--	5001438003B0B70E	QLogic Corporation	HPAE312A	5.01.03	9.1.8.25
Reported Host Name	DL36-529					
OS	Windows (2008 R2 Enterprise)					
OS Patch	SP0.0					
Architecture	AMD64					
IP	192.168.153.47					
Multipath	Microsoft DSM (6.1.7600.16385)					
Cluster Software	--					

The Host Explorer collects the following information and sends it to the storage server:

- Host operating system and version.
- Fibre Channel and iSCSI HBA details.
- Multipath driver and current multipath configuration.
- Cluster configuration information.
- Device and path Information.

Paths

WWN	Vendor	Model	Firmware Version	Driver Version	Speed	Multiple Attach
5001438003B0B70E	QLogic Corporation	HPAE312A	5.01.03	9.1.8.25	4 Gbps	Yes
5001438003B0B70C	QLogic Corporation	HPAE312A	5.01.03	9.1.8.25	4 Gbps	Yes



Host Persona

Why Host Personas?

- ▶ Different operating systems require slightly different behaviors, both from the port level and the SCSI layer
- ▶ Previously, there was port persona which combined port and SCSI layer behavior capabilities
- ▶ Host persona splits this into SCSI layer behaviors (host persona capabilities) and port behavior settings.



Host Personas

Advantages

- ▶ Purchase fewer and configure fewer ports for host connectivity
- ▶ Simplifies connecting hosts to the storage server and reduces management costs related to complex host connections
- ▶ Single StorServ port will support multiple operating systems accessing it



Host Personas

- ▶ Multiple host types can share a single StorServ Storage Server port
- ▶ Each host initiator gets its unique FC/SCSI level behavior
- ▶ Conversion from port personas to host personas will occur automatically during the upgrade process



Host persona management

- ▶ A host persona allows hosts that are connected to FC, iSCSI, or FCoE ports on the system to deviate from the default host behavior.
- ▶
- ▶ By assigning a persona to a host, multiple host types that require distinct customized responses can share a single system port.
- ▶ For example, hosts that run Microsoft Windows, Linux, and AIX operating systems can all connect to the same system port.
- ▶ Assigning a persona to a host simplifies connecting hosts to the system and reduces management costs related to complex host connection.



Host Persona Numbers

Operating System	Persona Number	Persona Name
Windows 2008 (R2)	1	Generic
Windows 2008 (non-R2)	2	Generic-ALUA
Windows 2003 (All versions)	1	Generic
VMWare (vSphere & ESX)	6	Generic-legacy
HP-UX	7	HPUX-legacy
AIX	8	AIX-legacy
Oracle Enterprise Linux	1	Generic

A host device must use either iSCSI or Fibre Channel connections. Mixed ports on a single device is not supported.

Active and inactive hosts

- ▶ An **active host** is a host that is connected to a system port and recognized by the HPE 3PAR OS. Under normal operation, an active host may have a number of volumes exported to it and therefore the host has access to those volumes.
- ▶ An **inactive host** is a host that is known to the HPE 3PAR OS but is not recognized as being connected to any system port at the moment.
- ▶ This inactivity may be because the host is disconnected from the system port, because the host is offline, or because of an error condition such as link failure.



Host Persona—Output Examples

```
root@xnodeed53:S420# showhost
```

Id	Name	Persona	-----WWN/iSCSI_Name-----	Port
0	sqa-sunu60-04	Generic-ALUA	210000E08B028F2F	---
			210100E08B228F2F	---
1	ibm520-13	AIX-legacy	10000000C97B142E	0:0:2
			10000000C97B142F	0:0:2
			10000000C97B142E	1:0:2
			10000000C97B142F	1:0:2
2	hprx2660-01	HPUX-legacy	500110A00011A998	0:0:2
			500110A00011A998	1:0:2
4	pe850-04-iscsi	Generic-legacy	iqn.1987-05.com.cisco:01.c2f3dffc24f9	
5	pe850-02	Generic-ALUA	210100E08B257D21	0:0:2
			210000E08B057D21	0:0:2
			210000E08B057D21	1:0:2
			210100E08B257D21	1:0:2
6	pe850-04	Generic-legacy	210000E08B041B50	0:0:2
			210000E08B041B50	1:0:2
--			210000E08B02DDFA	1:0:2
			210000E08B02DDFA	0:0:2
			210000E08B02DAFA	1:1:1



Host Persona—Output Examples

```
root@xnodeed53:S420# showhost -persona
```

Id	Name	Persona_Id	Persona_Name	Persona_Caps
0	sqa-sunu60-04	2	Generic-ALUA	UARepLun, RTPG, SESLun
1	ibm520-13	8	AIX-legacy	NACA
2	hprx2660-01	7	HPUX-legacy	VolSetAddr
4	pe850-04-iscsi	6	Generic-legacy	--
5	pe850-02	2	Generic-ALUA	UARepLun, RTPG, SESLun
6	pe850-04	6	Generic-legacy	--

```
root@xnodeed53:S420# showhost -listpersona
```

Persona_Id	Persona_Name	Persona_Caps
1	Generic	UARepLun, SESLun
2	Generic-ALUA	UARepLun, RTPG, SESLun
6	Generic-legacy	--
7	HPUX-legacy	VolSetAddr
8	AIX-legacy	NACA
9	EGENERA	SoftInq
10	ONTAP-legacy	SoftInq



Host Persona—Output Examples

```
root@xnodeed53:S420# showport
```

N:S:P	Mode	State	----Node_WWN----	-Port_WWN/HW_Addr-	Type
0:0:1	initiator	ready	2FF70002AC0001A4	20010002AC0001A4	free
0:0:2	target	ready	2FF70002AC0001A4	20020002AC0001A4	host
0:1:1	initiator	loss_sync	28110002AC0001A4	20110002AC0001A4	free
0:1:2	initiator	loss_sync	28120002AC0001A4	20120002AC0001A4	free
0:3:1	target	offline	-	00C0DD086664	iscsi
0:3:2	target	loss_sync	-	00C0DD086666	iscsi
0:4:1	initiator	ready	2FF70002AC0001A4	20410002AC0001A4	disk
0:4:2	initiator	ready	2FF70002AC0001A4	20420002AC0001A4	disk
0:5:1	peer	offline	-	0002B39B1FE6	rcip
1:0:1	target	ready	2FF70002AC0001A4	21010002AC0001A4	free
1:0:2	target	ready	2FF70002AC0001A4	21020002AC0001A4	host
1:1:1	target	ready	2FF70002AC0001A4	21110002AC0001A4	host
1:2:1	target	offline	-	00C0DD0779FD	iscsi
1:2:2	target	loss_sync	-	00C0DD0779FF	iscsi
1:4:1	initiator	ready	2FF70002AC0001A4	21410002AC0001A4	disk
1:4:2	initiator	ready	2FF70002AC0001A4	21420002AC0001A4	disk
1:5:1	peer	offline	-	0002B39B202A	rcip



Host Persona—Output Examples

```
root@xnoded53:S420# showport -par
```

N:S:P	Connmode	ConnType	CfgRate	MaxRate	Class2	UniqNodeWwn	VCN	IntCoal
0:0:1	disk	loop	auto	4Gbps	disabled	disabled	disabled	enabled
0:0:2	host	point	auto	4Gbps	disabled	disabled	disabled	enabled
0:1:1	disk	loop	auto	2Gbps	disabled	disabled	disabled	enabled
0:1:2	disk	loop	auto	2Gbps	disabled	disabled	disabled	enabled
0:4:1	disk	loop	auto	2Gbps	disabled	disabled	disabled	enabled
0:4:2	disk	loop	auto	2Gbps	disabled	disabled	disabled	enabled
1:0:1	host	loop	auto	2Gbps	disabled	disabled	disabled	enabled
1:0:2	host	point	auto	2Gbps	disabled	disabled	disabled	enabled
1:1:1	host	loop	auto	2Gbps	disabled	disabled	disabled	enabled
1:4:1	disk	loop	auto	2Gbps	disabled	disabled	disabled	enabled
1:4:2	disk	loop	auto	2Gbps	disabled	disabled	disabled	enabled

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↓

NOTE: Disk connections always use loop mode

```
root@xnodee3a3:~# controlport config disk 0:0:1
```

```
Are you sure you want to run controlport config disk on port 0:0:1 (connection type loop)?  
select q=quit y=yes n=no: y
```



Setting Personas—createhost and sethost

```
root@xnodee3a3:~# createhost sqa-sunu60-04 210100E08B228F2F 210000E08B028F2F
Setting default host persona 1 (Generic)
```

```
root@xnodee3a3:~# createhost -persona 1 sqa-sunu60-04 210100E08B228F2F 210000E08B028F2F
root@xnodee3a3:~#
```

```
root@xnodee3a3:~# sethost -persona 6 sqa-sunu60-04
root@xnodee3a3:~#
```



HPE 3par File Persona

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HPE 3PAR File Persona

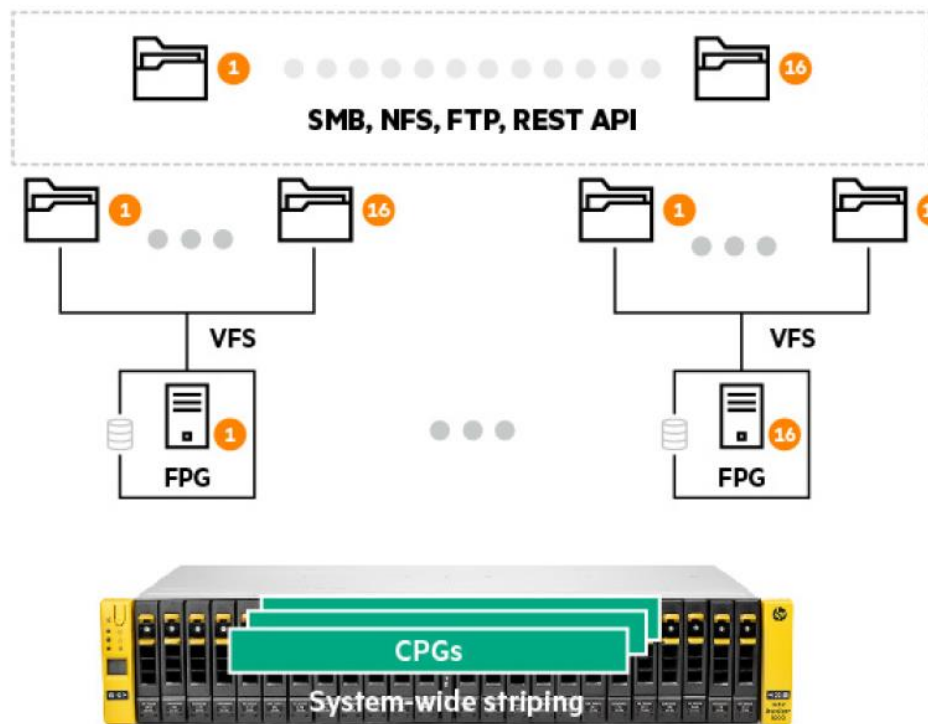
- ▶ With HPE 3PAR File Persona, you can create a converged storage solution with block and file storage services. This solution delivers tightly integrated, converged storage for provisioning.
- ▶ Both block volumes for server workloads, and file and object shares for client workloads such as home directory consolidation, can be provisioned.
- ▶ You must have network interface cards that support the File Persona software installed on the StoreServ to use the File Persona software.



-
- ▶ There are many advantages to running the HPE 3PAR File Persona software on a StoreServ:
 - ▶ Block and file services can use a single pool of thinly provisioned storage.
 - ▶ Autonomic data management services are applied to block and file data.
 - ▶ Resilient Mesh-Active architecture.
 - ▶ Leverages the advanced data services of HPE 3PAR OS.



File Persona Architecture



File Share (for example, "home")

- Share permissions

File Store (for example, "sales")

- Snapshot policies, capacity quotas, and customized virus scanning policies
- Entity for up to 1,024 snapshots

Virtual file Server (for example, "enterprise.hpe.com")

- Virtual IP interfaces
- User/group quotas and default virus scanning policy

File provisioning group (for example, "fpg1")

- Instance of HPE Adaptive File System transparently built from virtual volumes
- Entity for replication and disaster recovery

RESILIENCY AND HIGH AVAILABILITY

- ▶ File Persona uses a mission-critical, proven, 64-bit journaling file system that has been optimized for high metadata-driven workloads such as home directory consolidation and corporate and group shares.
- ▶ File Persona benefits from the inherited HPE 3PAR StoreServ resiliency. In the event of a node failover, the necessary File Persona objects will fail over to the other node in the node pair.
- ▶ Depending on the protocol, the failovers are transparent to the users.

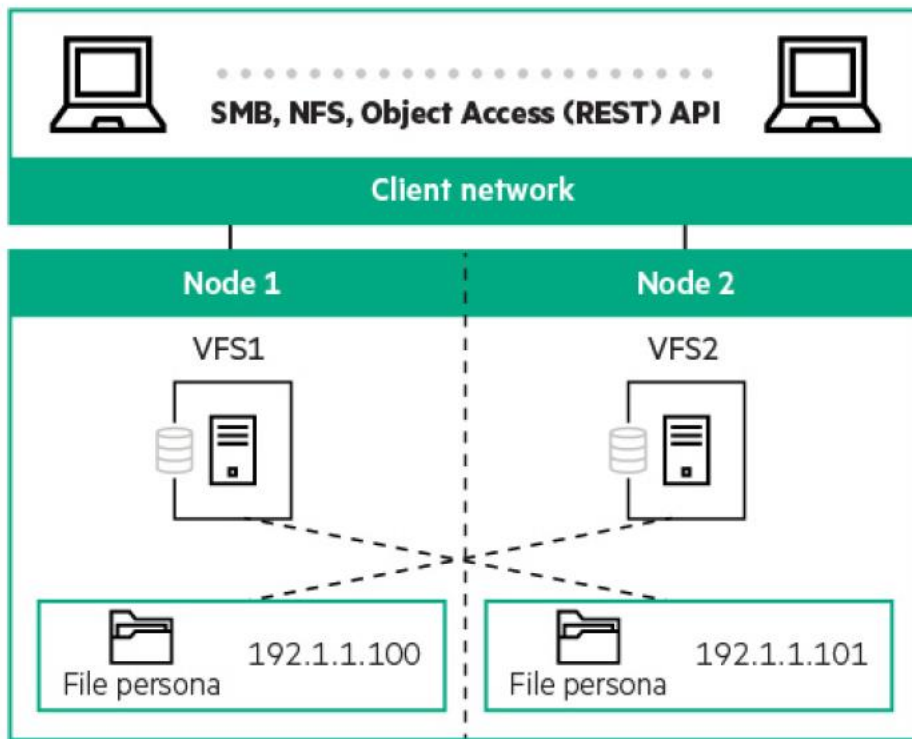


NETWORKING

- ▶ File Persona requires one of the following components to be installed in an available PCI slot on each node pair:
- ▶ Four-port 1GbE NIC
- ▶ Two-port 10GbE NIC
- ▶ Four-port combo card with:
 - ▶ – 2 x 16Gb Fibre Channel + 2 x 10GbE ports
 - ▶ – 2 x 10GbE iSCSI + 2 x 10GbE ports, available on HPE 3PAR StoreServ 8000 series only



Network Connectivity



File Persona supports antivirus scan policies to control scanning as well as on-access (real-time) and on-demand scanning. For redundancy and improved throughput performance, virus scanning can be configured with multiple antivirus scan servers. Scanned file information is persisted to avoid running redundant scans and wasting valuable resources.



QUOTA MANAGEMENT

- ▶ Quota management provides better control and planning for data growth, thus reducing the business cost for data backups and archiving. Furthermore, quotas balance resource utilization and help to ensure appropriate usage.
- ▶ Quotas can be combined with alerts, logs, and reporting events to maintain records. They are essential for organizations that implement a chargeback model in their environment.
- ▶ You can configure quotas used in File Persona with a hard threshold limit, which is immediately enforced after being exceeded (that is, users cannot write any more after the hard limit is reached), or a soft threshold limit.
- ▶ When a soft threshold limit is reached, a grace period (seven days by default) begins in which continued writes are allowed.



General

System ebc_7450c10

Virtual file server vfs1

Grace Time

Capacity grace time 10080 Minutes

File grace time 10080 Minutes

Quotas

Search



All statuses All types

0 quotas

	Name	Type	ID	Current Capacity (MiB)	Current Files	Soft Limit Capacity (MiB)	Hard Limit Capacity (MiB)	Soft Limit (Files)	Hard Limit (Files)
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No data available in table

Add

Remove

FILE LOCK FOR DATA IMMUTABILITY AND RETENTION

- ▶ Protecting critical data from accidental deletion or malicious alteration is a key requirement for most organizations today.
- ▶ To support this, File Persona offers File Lock for data immutability and retention.
- ▶ File Lock operates in two modes:
 - ▶ Enterprise mode and Compliance mode.



HPE 3PAR Virtual Lock

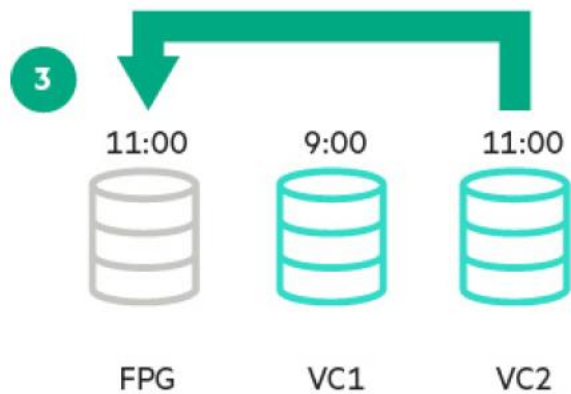
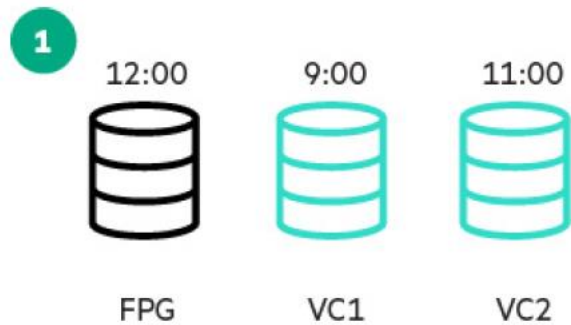
- ▶ HPE 3PAR Virtual Lock is an optional feature that enforces the retention period of any volume or copy of a volume.
- ▶ Locking a volume prevents the volume from being deleted, intentionally or unintentionally, before the retention period elapses.
- ▶ You can use HPE 3PAR Virtual Lock to specify the retention period for each volume or copy of a volume.



Administrator-driven recovery

- ▶ In addition to user-driven recovery, HPE 3PAR File Persona also allows the storage administrator to recover the entire FPG using HPE 3PAR Virtual Copy technology, which provides a crash-consistent point-in-time snapshot of the entire FPG.
- ▶ This is useful for FPG recovery, in the event of a file system problem. The FPG is rolled back to the previous point in time at which the Virtual Copy was created.
- ▶ This recovery occurs offline, because the FPG has to be unmounted before the recovery starts.





Replication and disaster recovery

- ▶ For File Persona replication, HPE 3PAR Remote Copy software is used the same for File Persona VVs and for Block Persona VVs, supporting Synchronous, Asynchronous Periodic, and Synchronous Long Distance modes.
- ▶ All VVs in an FPG must be in a single Remote Copy group.
- ▶ File Persona supports 1:1, many-to-one (M:1), and one-to-many (1:N) replication topologies for failover purposes only—not for distribution purposes as a read-only target.
- ▶ As a prerequisite for disaster recovery configuration for File Persona, the node networking, DNS configuration, Active Directory configuration, antivirus integration, scheduled tasks, and other pertinent features have to be set up manually on the remote array.



Traditional backup

- ▶ File Persona supports network share-based backup over SMB or NFS protocols and NDMP over iSCSI-based backup. Share-based backups can use any ISV that supports backups from file shares.
- ▶ File Persona supports the following software vendors for NDMP-based backup and restore functions:
 - ▶ Micro Focus Data Protector
 - ▶ Veritas NetBackup
 - ▶ Commvault Simpana
 - ▶ Dell EMC NetWorker





GENERAL	BLOCK PERSONA	FILE PERSONA	STORAGE OPTIMIZATION	DATA PROTECTION	STORAGE SYSTEMS
Dashboard	Hosts	File Shares	Adaptive Flash Cache	Remote Copy Configurations	Systems
Activity	Host Sets	File Stores	Adaptive Optimization	Remote Copy Groups	Controller Nodes
Schedules	Virtual Volumes	Virtual File Servers	Priority Optimization	RMC Instances	Ports
Settings	App Volume Sets	File Provisioning Groups		Restore Points	Drive Enclosures
	Virtual Volume Sets				Physical Drives
	Common Provisioning Groups	File Persona Configuration			
	Policies				
SECURITY	VMWARE				
Users	Storage Containers				
LDAP	Virtual Machines				
Roles					
Connections					
Domains					



Configure Node Pair



Pair name 0-1

State Configured

NIC Pairs (N:S:P) 1 NIC Pairs

Node 0 ▲ Node 1

0:3:* 1:3:* ✕

Add

1 added

Node 0 IP address 192.168.206.220

Node 1 IP address 192.168.206.22

Supply the IP address used to identify this node to authentication services.

CPG System default for file

Select

optional



Changed: Node 0 IP address to "192.168.206.220"

OK

Cancel

dxs-3par cli%

dxs-3par cli% startfs -enable 0,1

Cannot install File Persona. No memory reserved for File Persona on all nodes. Please contact HPE Support.

dxs-3par cli% █



File Persona Configuration 1

Status ▾

All ▾

File Persona State ▾

All ▾



⚠️ dxc-3par

Overview ▾



● System ▲ File Persona State

▲ dxc-3par Licensed

General

File provisioning groups [Create file provisioning group](#)Virtual file servers [Create virtual file server](#)

Authentication providers 0

Antivirus servers 0

Health

State ⚠️ Warning

State description
Licensed for File Persona
File Persona not running

Controller Nodes

●	Node ID ▲	File Persona State	Active	InCluster	Version	Node IP Address
■	0	Disabled	No	No		Not set
■	1	Disabled	No	No		Not set

Storage System

☐ Advanced options

Storage system dxc-3par

Compliance WORM feature is not supported for selected system.

Node Pairs

Name	▲ State	1st IP Address	2nd IP Address
0-1	Unconfigured	not set	not set 

Network Settings

Netmask

Gateway

Enter a netmask, (i.e. 255.255.255.0). The netmask is a global setting and will be applied to all IP addresses used by the file persona and virtual file servers.

All systems ▾

Health

State

⚠️ Warning

State description

Licensed for File Persona
File Persona not running

Actions ▾

Configure file persona

Resume file persona node

Pause file persona node

Delete file persona node pair

Manage protocol settings

Delete LDAP configuration

Leave active directory

Update virus definition

Manage local users

Manage local groups

Manage user mappings

Export user mappings

Recover file provisioning groups

Create virtual file server

Node IP Address

Not set

Not set

Network Settings

1 (Active/Standby)

☒ Advanced options

DNS Settings

- ▶ DNS Servers optional
- ▶ Search Suffixes optional

Authentication Settings

▼ Provider Order optional

1st: Active Directory

2nd: Local Users & Groups

3rd: Not used

The order the system should attempt to resolve user credentials in.

- ▶ Local Users optional
- ▶ Local Groups optional
- ▶ NFSv4 mapid domain optional
- ▶ LDAP Configuration Settings optional
- ▼ Active Directory Settings optional

Configure

Cancel

Create File Snapshot

General

?

General

System

Show Taylor all machines

Virtual file server

vfs1

File store

home

Tag

LocalSnap

Retention

4

Enter the number of snapshots to retain for this file store. Once this number is reached, the oldest snapshot is deleted when a new one is created. The maximum number of snapshot files allowed per file store is 1024.

Schedule

Schedule name

DailySnap

Alert

Generate if task failed

Schedule pattern

Daily

Starting at

06:00

 6

Changed: Schedule name to "DailySnap"

OK

Cancel

Thank you . . .

